

JHANVI U SUNDAR

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Education

Ramaiah Institute of Technology | Bangalore, Karnataka

B.E. in Artificial Intelligence & Data Science — CGPA: 8.69 | 2023 – Present

Vidya Niketan School ISC — 77.2% | 2023

Sophia High School ICSE — 96.3% | 2021

Experience

Software Development Intern, Leverage IT

Summer 2025

- Processed and transformed large-scale JSON and streaming datasets into ML-ready formats through data cleaning, normalization, and preprocessing pipelines.
- Engineered structured training and testing datasets by resolving data quality issues and standardizing heterogeneous data sources for downstream model development.
- Leveraged Azure Blob Storage to manage cloud-hosted datasets and support efficient data ingestion workflows for machine learning applications.
- Collaborated with senior engineers on machine learning model training and validation workflows using GPU-enabled cloud infrastructure.

Tech stack: Python, Azure Blob Storage, Pandas, NumPy

Projects

PathWise – Context-Aware Multi-Agent AI Routing System

3rd Place – Google DeepStation x RIT Hackathon 2025

- Architected a multi-agent AI routing platform integrating Google ADK, Vertex AI, Gemini 2.5 Flash, Google Maps APIs, and OpenStreetMap data to generate personalized route recommendations.
- Developed 5+ specialized AI agents for route orchestration, weather intelligence, scenic analysis, vehicle compatibility assessment, and safety evaluation.
- Engineered a computer-vision-based road width estimation pipeline using satellite imagery, enabling passability assessment with confidence-based scoring.
- Designed a dynamic route-ranking engine incorporating 10+ environmental and geographic factors including weather, landmarks, seasonality, terrain, and visual quality.

Tech stack: Python, Google ADK, Vertex AI, Gemini 2.5 Flash, Google Maps APIs, OpenStreetMap, OpenCV

SepsisSense – Clinical Decision Support & Pathway Discovery Platform

- Engineered a hybrid Monte Carlo and Ant Colony Optimization (MC²-ACO) framework to analyze 24,000 clinical observation windows and discover patient deterioration pathways leading to sepsis.
- Developed an end-to-end clinical analytics platform with FastAPI and React for interactive cohort analysis and pathway visualization.
- Achieved 87.17% AUROC and 82.22% sensitivity in identifying high-risk sepsis progression patterns across clinical cohorts.
- Validated pathway robustness using bootstrap testing across 500 random seeds, achieving 100% stability for Warm and Cryptic Sepsis cohorts.

Tech stack: Python, FastAPI, React, NumPy, Pandas

Smart Waste Management System (AI-IoT)

Ongoing

- Designed an AI-powered waste monitoring and sorting system integrating computer vision, IoT sensors, and embedded systems to improve sustainability and operational efficiency.
- Proposed a scalable architecture for automated waste classification, collection monitoring, and real-time analytics.
- Presented the project proposal to department leadership and secured institutional funding approval for full project development.
- Currently developing the AI and sensor integration pipeline for automated waste identification and monitoring.

Tech stack: Python, IoT Sensors, Computer Vision, Embedded Systems

Plankton Pi – Automated Marine Organism Classification

- Developed a machine learning pipeline for automated detection, counting, and classification of microscopic marine organisms from microscopy images.
- Integrated YOLOv8-seg and EfficientNet models to support high-volume environmental monitoring and marine biodiversity analysis.
- Designed the software prototype for rapid organism characterization, reducing manual inspection requirements for researchers.
- Continuing development of a Raspberry Pi-based edge deployment system with focus on inference optimization and offline operation.

Tech stack: Python, YOLOv8-Seg, EfficientNet, OpenCV

Skills

Languages: Python, Java, C

Frameworks & Libraries: FastAPI, Flask, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, YOLOv8, EfficientNet

AI & Machine Learning: Computer Vision, Object Detection, Image Segmentation, Multi-Agent Systems, LLM Applications, Route Optimization, Satellite Image Processing

Cloud & Tools: Vertex AI, Google ADK, Google Cloud Run, Azure Blob Storage, TensorRT, Git Hub

Achievements

- 3rd Place, Google x Deepstation xRIT Hackathon 2025 for developing PathWise, a context-aware multi-agent AI routing platform leveraging multi-agent AI, computer vision, and geospatial intelligence.
- Google Student Ambassador.
- Secured institutional funding approval for Smart Waste Management, an AI-IoT sustainability initiative presented to department leadership.
- Athletics Team Captain with multiple inter-school medals.